

Assignment-1

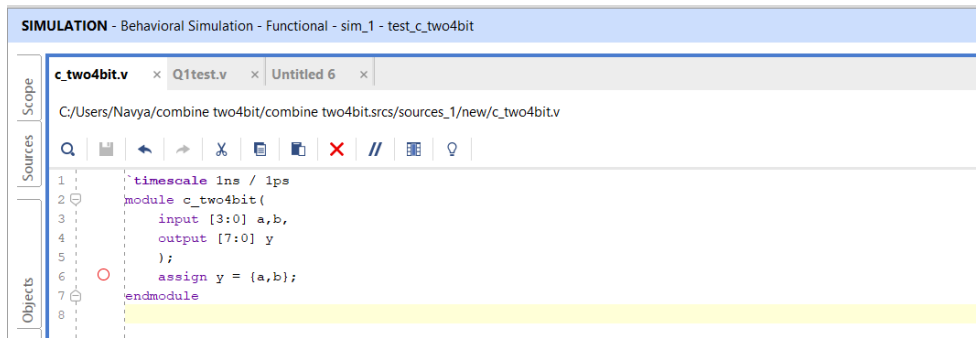
DIGITAL SYSTEMS LAB

DESIGN AND SIMULATION USING VERILOG

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QUESTION 1: COMBINE TWO 4-BIT NUMBERS

Design Code



The screenshot shows the Verilog code for a module named `c_two4bit`. The code is as follows:

```
1 timescale 1ns / 1ps
2 module c_two4bit(
3     input [3:0] a,b,
4     output [7:0] y
5 );
6     assign y = {a,b};
7 endmodule
```

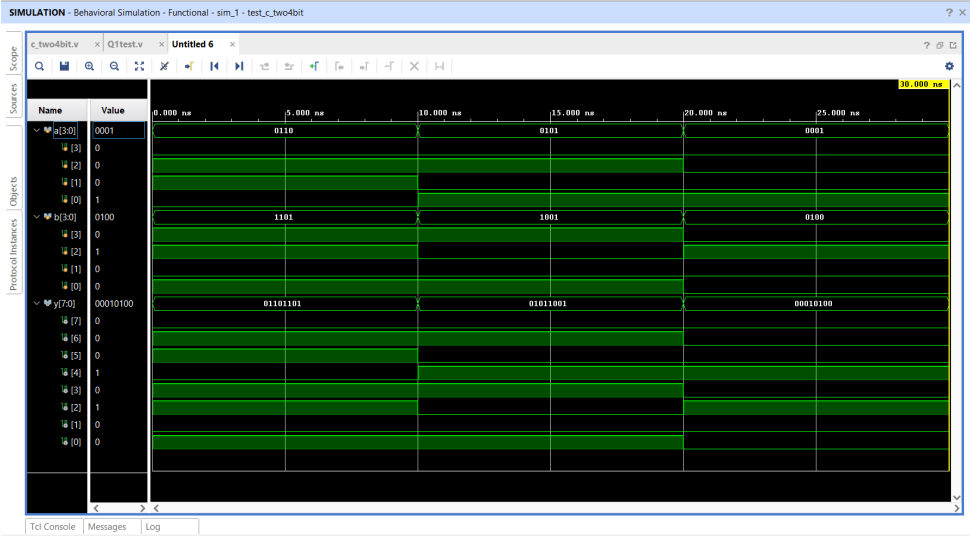
Testbench Code



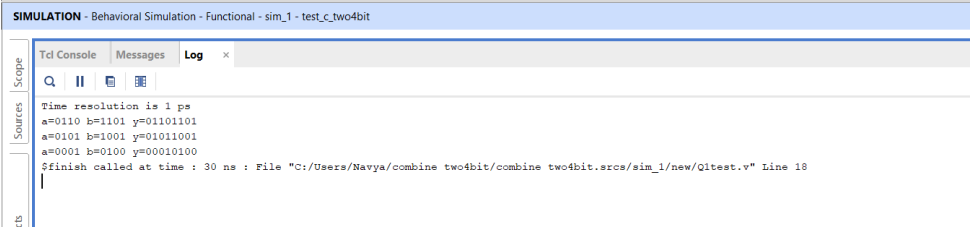
The screenshot shows the Verilog testbench code for the `c_two4bit` module. The code is as follows:

```
1 timescale 1ns / 1ps
2 module test_c_two4bit;
3     reg [3:0] a, b;
4     wire [7:0] y;
5
6     c_two4bit uut (.a(a), .b(b), .y(y));
7
8     initial begin
9         a = 4'b0110; b = 4'b1101; #10;
10        $display("a=tb b=tb y=tb", a, b, y);
11
12        a = 4'b0101; b = 4'b1001; #10;
13        $display("a=tb b=tb y=tb", a, b, y);
14
15        a = 4'b0001; b = 4'b0100; #10;
16        $display("a=tb b=tb y=tb", a, b, y);
17
18        $finish;
19    end
20 endmodule
```

Simulation Waveform



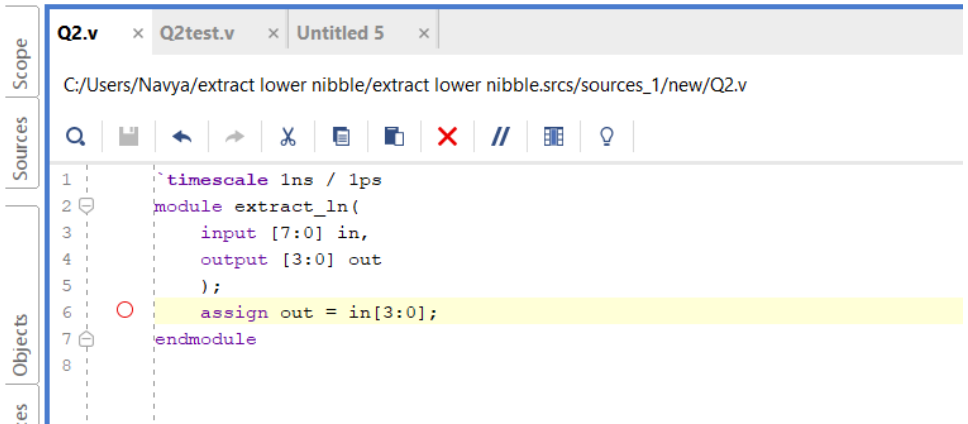
Output



QUESTION 2: EXTRACT LOWER NIBBLE

Design Code

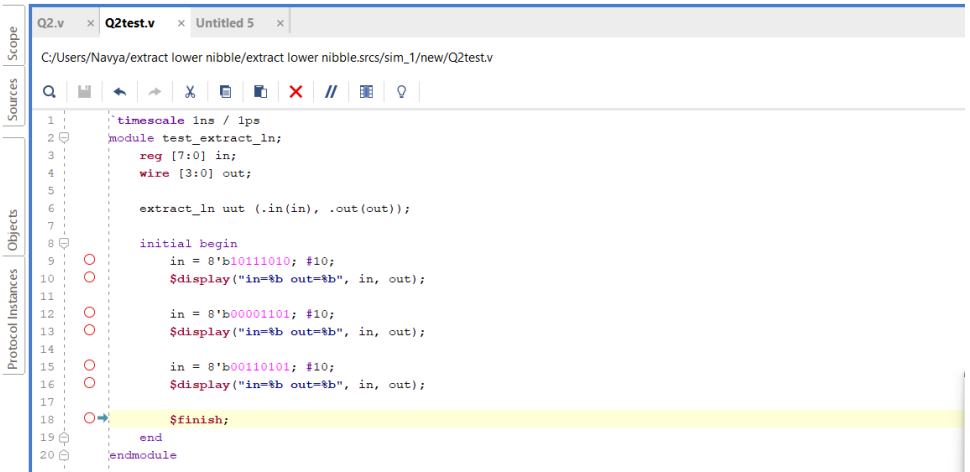
SIMULATION - Behavioral Simulation - Functional - sim_1 - test_extract_ln



```
1 timescale 1ns / 1ps
2 module extract_ln(
3     input [7:0] in,
4     output [3:0] out
5 );
6     assign out = in[3:0];
7 endmodule
8
```

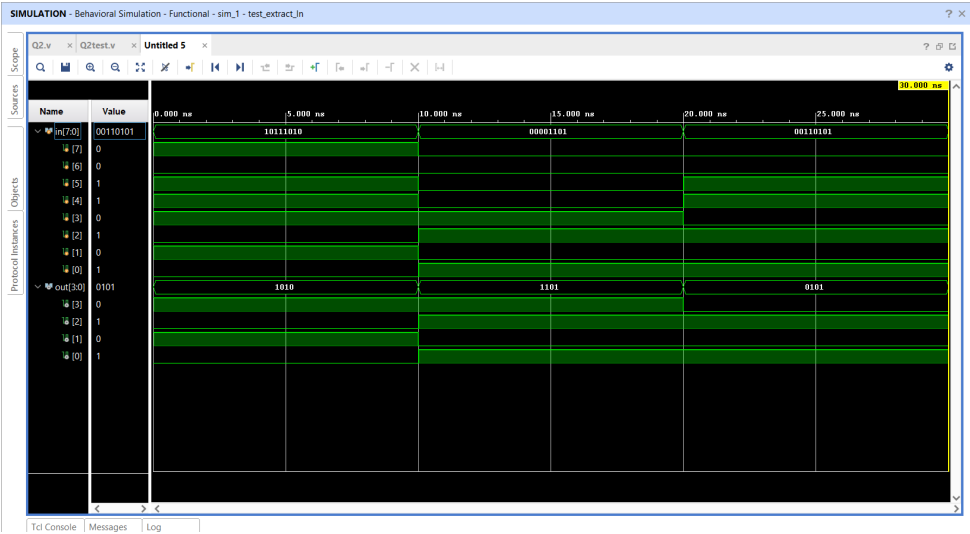
Testbench Code

SIMULATION - Behavioral Simulation - Functional - sim_1 - test_extract_ln

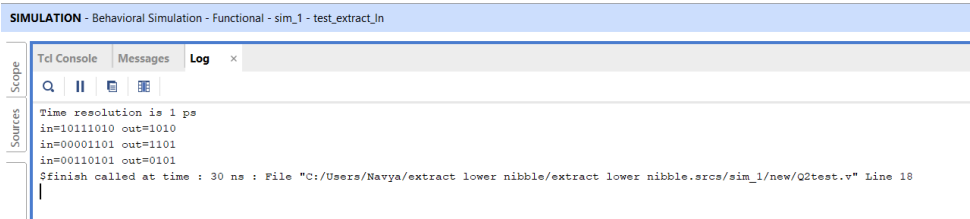


```
1 timescale 1ns / 1ps
2 module test_extract_ln;
3     reg [7:0] in;
4     wire [3:0] out;
5
6     extract_ln uut (.in(in), .out(out));
7
8     initial begin
9         in = 8'b10111010; #10;
10        $display("in=%b out=%b", in, out);
11
12        in = 8'b00001101; #10;
13        $display("in=%b out=%b", in, out);
14
15        in = 8'b00110101; #10;
16        $display("in=%b out=%b", in, out);
17
18        $finish;
19    end
20 endmodule
```

Simulation Waveform

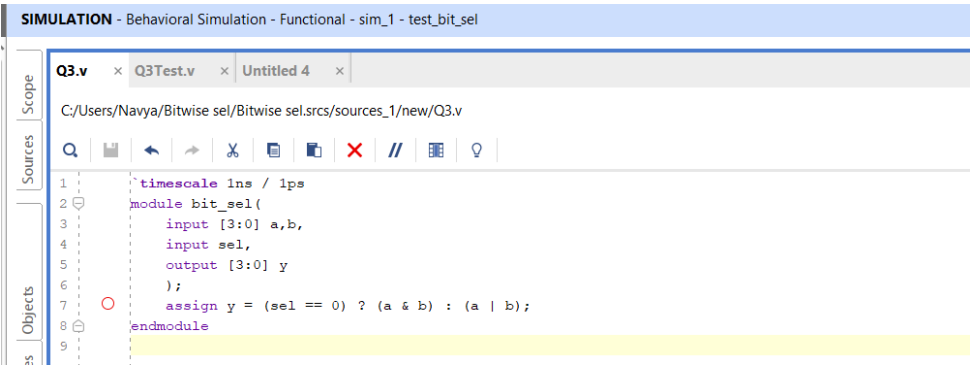


Output



QUESTION 3: BITWISE AND OR OR SELECTOR

Design Code



Testbench Code

SIMULATION - Behavioral Simulation - Functional - sim_1 - test_bit_sel

Q3.v x Q3Test.v x Untitled 4 x

C:/Users/Navya/Bitwise sel/Bitwise sel.srcs/sim_1/new/Q3Test.v

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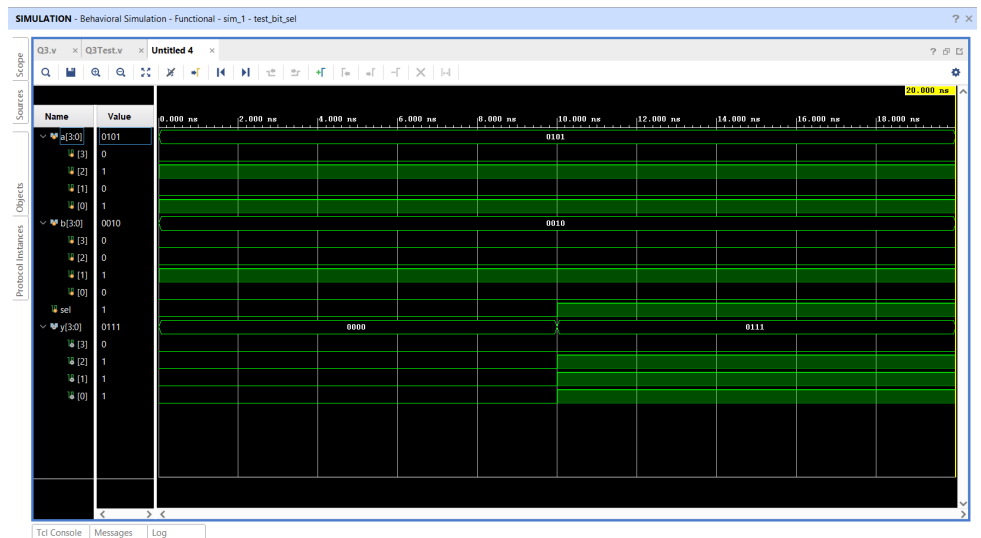
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ScopeSourcesObjectsProtocol Instances

```
1 timescale 1ns / 1ps
2 module test_bit_sel;
3     reg [3:0] a, b;
4     reg sel;
5     wire [3:0] y;
6
7     bit_sel uut (.a(a), .b(b), .sel(sel), .y(y));
8
9     initial begin
10         a = 4'b0010; b = 4'b0010; sel = 0; #10;
11         $display("sel=%b a=%b b=%b y=%b", sel, a, b, y);
12
13         sel = 1; #10;
14         $display("sel=%b a=%b b=%b y=%b", sel, a, b, y);
15
16         $finish;
17     end
18 endmodule
```

Simulation Waveform



Output

SIMULATION - Behavioral Simulation - Functional - sim_1 - test_bit_sel

Tcl ConsoleMessagesLog

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Time resolution is 1 ps
sel=0 a=0101 b=0010 y=0000
sel=1 a=0101 b=0010 y=0111
\$finish called at time : 20 ns : File "C:/Users/Navya/Bitwise sel/Bitwise sel.srcs/sim_1/new/Q3Test.v" Line 16

QUESTION 4:SIMPLE EVEN PARITY GENERATOR

Design Code

SIMULATION - Behavioral Simulation - Functional - sim_1 - test_even_parity

Q4.v *Q4Test.vUntitled 1

C:/Users/Navya/even parity/even parity.srcs/sources_1/new/Q4.v

Q

```
1 timescale 1ns / 1ps
2 module even_parity(
3     input [3:0] data,
4     output parity
5 );
6     assign parity = data[0] ^ data[1] ^ data[2] ^ data[3];
7 endmodule
```

Testbench Code

PROJECT MANAGER - even parity

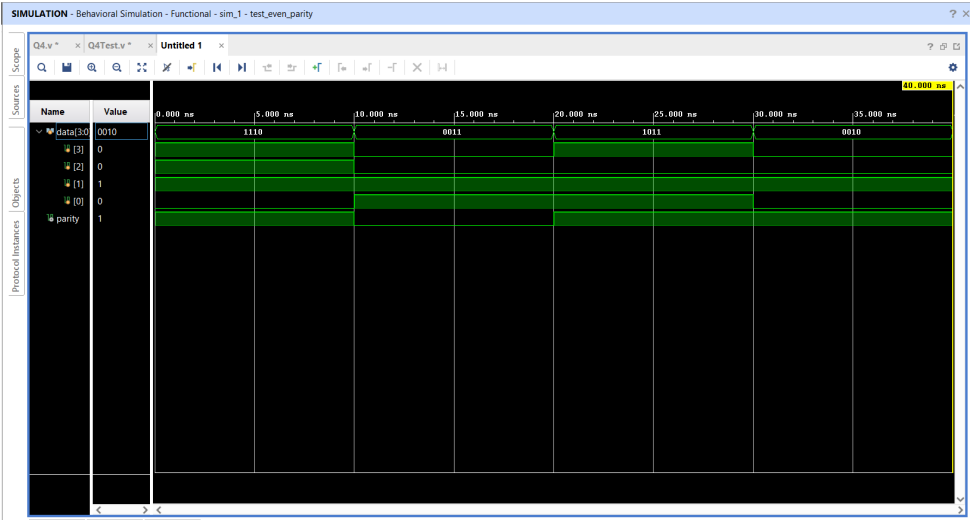
Project SummaryQ4Test.v

C:/Users/Navya/even parity/even parity.srcs/sim_1/new/Q4Test.v

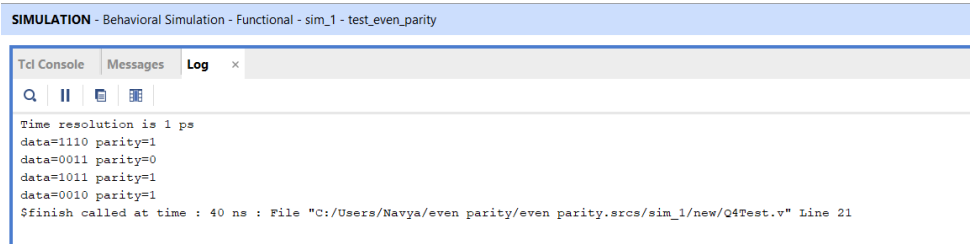
Q

```
1 timescale 1ns / 1ps
2 module test_even_parity;
3     reg [3:0] data;
4     wire parity;
5
6     even_parity uut (.data(data), .parity(parity));
7
8     initial begin
9         data = 4'b1110; #10;
10        $display("data=%b parity=%b", data, parity);
11
12        data = 4'b0011; #10;
13        $display("data=%b parity=%b", data, parity);
14
15        data = 4'b1011; #10;
16        $display("data=%b parity=%b", data, parity);
17
18        data = 4'b0010; #10;
19        $display("data=%b parity=%b", data, parity);
20
21        $finish;
22    end
23 endmodule
```

Simulation Waveform

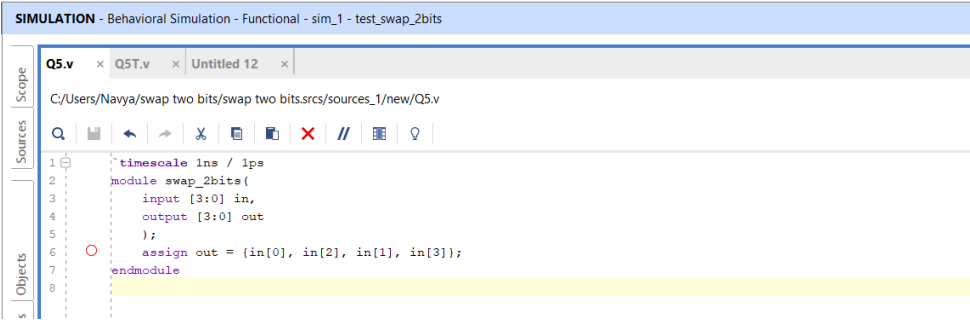


Output



QUESTION 5:SWAP TWO BITS

Design Code



Testbench Code

SIMULATION - Behavioral Simulation - Functional - sim_1 - test_swap_2bits

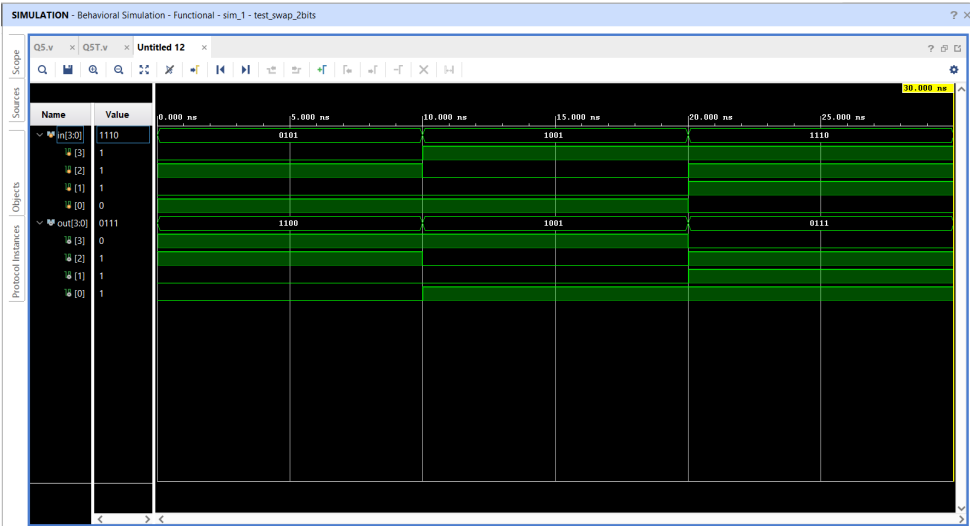
Q5.v x Q5T.v x Untitled 12 x

C:/Users/Navya/swap two bits/swap two bits.srscs/sim_1/new/Q5T.v

ScopeSourcesObjectsProtocol Instances

```
1 timescale 1ns / 1ps
2 module test_swap_2bits;
3     reg [3:0] in;
4     wire [3:0] out;
5
6     swap_2bits uut (.in(in), .out(out));
7
8     initial begin
9         in = 4'b0101; #10;
10        $display("in=%b out=%b", in, out);
11
12        in = 4'b1001; #10;
13        $display("in=%b out=%b", in, out);
14
15        in = 4'b1110; #10;
16        $display("in=%b out=%b", in, out);
17
18        $finish;
19    end
20 endmodule
21
```

Simulation Waveform



Output

SIMULATION - Behavioral Simulation - Functional - sim_1 - test_swap_2bits

Tcl ConsoleMessagesLog x

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Time resolution is 1 ps
in=0101 out=1100
in=1001 out=1001
in=1110 out=0111
\$finish called at time : 30 ns : File "C:/Users/Navya/swap two bits/swap two bits.srcs/sim_1/new/Q5T.v" Line 18